LESSON PLAN

BASIC RADIO & AVIONICS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Network, LAN, WAN and Network Topology
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Network, LAN, WAN and Network Topology.
5. Specific Objectives	22.1.2.1 Trainee will able to explain existing IT networking system in IN

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of onboard IT architecture diagrams; Safety: No live electrical connections; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a network?
(b) Can a computer operate without a network connection?
(c) Name one device used to connect computers in a local area.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and scope of Computer Network • Classification of networks by geographical area • Characteristics of LAN and WAN systems • Role of IN IT networking infrastructure • Comparison of LAN and WAN protocols	L & Ds	PPT, IB	15 min
(b)	• Star topology structure and advantages • Bus topology implementation and limitations • Ring topology data flow mechanisms • Mesh topology redundancy features • Hybrid topology combinations in modern systems	L & Ds	PPT, PDF	20 min
(c)	• Selection criteria for network topology • Impact of topology on network performance • Troubleshooting common topology failures • Future trends in aviation network design • Security implications of different topologies	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a LAN and a WAN in terms of coverage.
(b) Why might a Star topology be preferred over a Bus topology in an aircraft cabin?
(c) Describe one advantage of a Mesh topology regarding fault tolerance.

12. Home Assignment and Follow Up	Prepare a brief note on how the existing IT networking system in India supports aviation data exchange, referencing concepts learned in this lesson to bridge to LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min